Requirement Specification Document (RSD)

Project Title

Paul's International E-Commerce Platform Integrated with Inventory Management System (IMS) for Multi-Country

Document Purpose

This document defines the business, functional, non-functional, technical, and delivery requirements for vendors to design, develop, deploy, and support a robust Inventory Management System (IMS) integrated with a full-featured E-Commerce website for a grocery retail business operating in two countries (India & Mexico).

The intent is to solicit comparable, technically sound, and commercially clear proposals from qualified vendors.

1. Business Overview

1.1 Business Model

- Grocery retail business
- Operations in two countries:
  - India
  - Mexico
- Each country has:
  - One or more warehouses
  - Multiple retail stores
- Business models supported:
  - Warehouse → Store → Customer
  - Warehouse → Customer (E-commerce)
  - Store → Customer (Walk-in & Online order fulfillment)

1.2 Key Business Goals

- Centralized, real-time inventory visibility
- Accurate stock, cost, margin, and profit tracking
- Country-wise and currency-wise financial reporting
- Scalable e-commerce platform usable in both India & Mexico
- Compliance with local tax and regulatory norms
- Reduced manual intervention and errors

2. Scope of Work (Vendor Responsibility)

The selected vendor shall be responsible for: 1. Requirement validation & solution design 2. Development of Inventory Management System (IMS) 3. Development of E-Commerce Website (India & Mexico) 4. Integration between IMS and E-Commerce 5. Payment gateway and courier integrations 6. Security, performance, and scalability 7. Deployment, documentation, and training 8. Post-go-live support and maintenance

3. Inventory Management System (IMS) – Functional Requirements

3.1 Master Data Management

3.1.1 Product Master

- SKU-based product management
- Support for:
    - Category / Sub-category
    - Brand
    - Unit of Measure (Kg, gm, L, pcs, pack)
    - Batch / Lot
    - Expiry date (mandatory for grocery items)
    - Barcode / QR code
- Country-specific pricing
- Multi-currency cost & selling price

3.1.2 Location Master

- Country
- Warehouse
- Store
- Each location must have:
    - Address
    - Time zone
    - Currency
    - Tax regime

3.1.3 Supplier & Vendor Master

- Supplier details
- Country of origin
- Currency
- Payment terms
- GST/VAT/Tax ID

3.1.4 Customer Master (Shared with E-Commerce)

- Customer profile
- Country-wise segmentation
- Purchase history
- Loyalty points (optional)

3.2 Inventory Management

3.2.1 Stock Tracking

- Real-time stock by:
  - Country
  - Warehouse
  - Store
- Available stock
- Reserved stock
- Damaged / expired stock
- Stock aging

3.2.2 Inventory Movements

- Goods Receipt (GRN)
- Warehouse-to-Store transfer
- Store-to-Store transfer
- Stock adjustments
- Returns (customer & supplier)

3.2.3 Expiry & Batch Management

- Batch-wise inventory
- FIFO / FEFO logic
- Expiry alerts (configurable days)
- Auto-blocking expired stock from sale

3.3 Sales & Billing

- POS sales (store sales)
- Online sales (from e-commerce)
- Invoice generation
- Country-specific tax handling:
  - India: GST
  - Mexico: IVA
- Multi-currency invoicing

3.4 Purchase Management

- Purchase Orders (PO)
- Supplier invoices
- GRN matching
- Cost price tracking
- Landed cost (optional)

3.5 Reporting & Analytics

3.5.1 Standard Reports (Mandatory)

Inventory Reports - Stock on hand - Low stock - Expiry report - Dead stock

Sales Reports - Daily / Weekly / Monthly / Annual sales - Country-wise sales - Store-wise sales - Product-wise sales

Profitability Reports - Gross margin - Net profit - Country-wise profit - Store-wise profit

Purchase Reports - Supplier-wise purchase - Cost variance

3.5.2 Report Features

- Export to Excel / PDF
- Date range filters
- Country / warehouse / store filters
- Dashboard with charts

4. E-Commerce Website Requirements

4.1 General

- Single platform supporting **India & Mexico**
- Country auto-detection or manual selection
- Language support:
    - English (mandatory)
    - English / Spanish (mandatory for Mexico, google translator can be used)
- Responsive design (mobile, tablet, desktop)

4.2 UI / UX

- Modern, clean, grocery-focused design
- Vendor may use **paid theme** (cost to be specified)
- Fast loading
- SEO-friendly structure

4.3 User Management & Security

- Customer registration
- Login options:
    - Email + password
    - Mobile number + OTP
- Secure credential storage
- GDPR-aligned customer data protection
- Role-based access (admin, staff, customer)

4.4 Product Catalog

- Category & sub-category
- Country-specific pricing
- Real-time stock visibility
- Product images & descriptions
- Offers & discounts

4.5 Cart & Checkout

- Persistent cart
- Guest checkout (optional)
- Address management
- Delivery slot selection
- Promo codes

4.6 Payment Gateway Integration

India - UPI - Credit / Debit cards - Net banking

Mexico - Cards - Local payment gateways

- Vendor to propose gateways
- Secure payment handling

4.7 Courier & Logistics Integration

- Multiple courier APIs
- Country-specific courier partners
- Auto shipment creation
- Live tracking
- Tracking link visible to customer

4.8 Notifications & Communication

- Order confirmation
- Shipping updates
- Invoice delivery:
    - Email
    - WhatsApp (mandatory)
- WhatsApp chat support integration

4.9 CMS & Admin Panel

- Banner management
- Product management
- Order management
- Customer management
- Content pages (About, FAQ, Policies)

5. Integration Requirements

- Real-time sync between:
    - IMS ↔ E-Commerce
- Stock update on:
    - Online sale
    - Store sale
    - Manual adjustment
- Single source of truth for inventory

6. Non-Functional Requirements

6.1 Performance

- System should handle:
  - ~1,000+ SKUs (should be scalable)
  - Concurrent users
- Page load < 3 seconds

6.2 Security

- HTTPS
- Encrypted credentials
- Secure APIs
- Role-based permissions

6.3 Scalability

- Ability to add new countries
- Ability to add new warehouses/stores

6.4 Availability

- 99.5% uptime
- Backup & disaster recovery

7. Technology Expectations (Vendor to Propose)

- Backend architecture
- Database design
- Cloud hosting (AWS / Azure / GCP)
- Mobile-friendly frontend
- API-based integrations

8. Deliverables

- Detailed Solution Architecture
- Functional System
- Source code ownership clarity
- Deployment setup
- Admin & user manuals
- Training sessions

9. Support & Maintenance

- Warranty period
- SLA for bug fixes
- Enhancement support

10. Vendor Proposal Expectations

Vendor must submit: - Technical approach - Project timeline - Team structure - Commercial pricing - Ongoing support cost - Assumptions & exclusions

11. Acceptance Criteria

- All modules implemented as per this document
- Successful UAT
- Stable production deployment

12. Wireframe Expectations

- Admin Dashboard- Global KPIs: Stock Value, Sales, Revenue, Profit
- Filters: Country, Warehouse, Store
- Alerts: Low Stock, Expiry, Pending Orders Inventory Screens
- Product Master (SKU, Batch, Expiry, Barcode)
- Warehouse-wise Stock View
- Stock Transfer Screens Reports & Analytics- Sales, Profit, Inventory, Tax Reports
- Export to PDF / Excel
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13. Data Migration Requirements

- Excel-based import templates
- Masters & opening stock migration
- Validation & reconciliation mandatory
- One dry run + final migration
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14. Legal & Compliance

- Data Ownership - All data belongs to business owner
- Intellectual Property - Custom source code provided to business owner
- Penalties & SLA - Delay & uptime penalties
- Compliance- India GST, DPDP Act- Mexico SAT, IVA